9.2 Two-Stream CNNs

- video can naturally be decomposed into spatial and temporal components
- the spatial part, in the form of individual frame appearance, carries information about scenes and objects depicted in the video
- the temporal part, in the form of motion across the frames, conveys the movement of the observer (the camera) and the objects
- the proposed video recognition architecture is devised accordingly, dividing it into two streams, as shown in Figure 3
- each stream is implemented using a deep CNN, softmax scores of which are combined by late fusion
- two fusion methods are considered: averaging and training a multi-class linear SVM on stacked ℓ_2-normalized softmax scores as features

9.2 Two-Stream CNNs

- **Spatial stream CNN** operates on individual video frames, effectively performing action recognition from still images
- the static appearance by itself is a useful clue, since some actions are strongly associated with particular objects
- in fact, action classification from still frames (the spatial recognition stream) is fairly competitive on its own
- since a spatial CNN is essentially an image classification architecture, the network can be pre-trained on a large image classification dataset, such as the ImageNet dataset
- next, we describe the temporal stream CNN, which exploits motion and significantly improves accuracy

9.2 Two-Stream CNNs

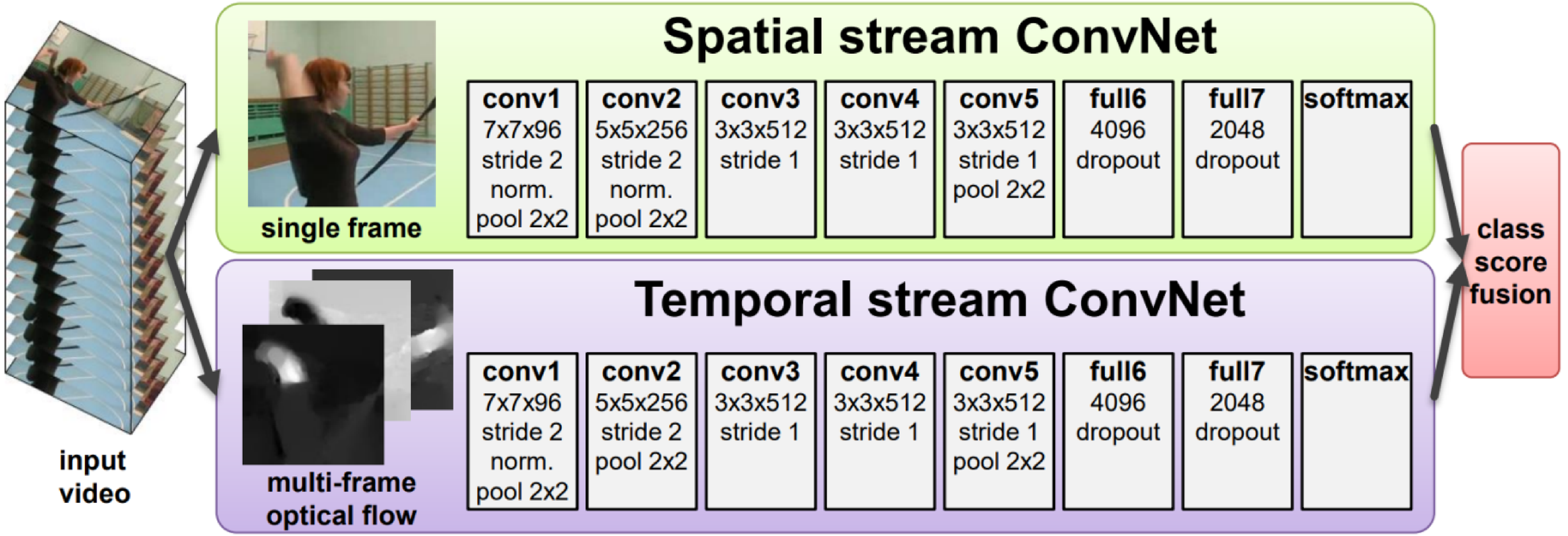


Figure 3: Two-stream architecture for video classification.

9.2 Two-Stream CNNs

- **Optical flow CNNs**: now, we describe a CNN model, which forms the temporal recognition stream of the architecture
- unlike the CNNs models, the input to the model is formed by stacking optical flow displacement fields between several consecutive frames
- such input explicitly describes the motion between video frames, which makes the recognition easier, as the network does not need to estimate motion implicitly
- several variations of the optical flow-based input are considered, which are described below

9.2 Two-Stream CNNs

- **Optical flow stacking**: a dense optical flow can be seen as a set of displacement vector fields $\boldsymbol{d}_t$ between the pairs of consecutive frames t and $t+1$
- by $\boldsymbol{d}_t(u, v)$ we denote the displacement vector at the point (u, v) in frame t, which moves the point to the corresponding point in the following frame $t+1$
- the horizontal and vertical components of the vector field, d_t^x and d_t^y, can be seen as image channels (shown in Figure 4), well suited to recognition using a convolutional network
- to represent the motion across a sequence of frames, the flow channels $d_t^{x,y}$ of L consecutive frames are stacked to form a total of $2L$ input channels
- more formally, let w and h be the width and height of a video; a CNN input volume $I_\tau \in \mathbb{R}^{w \times h \times 2L}$ for an arbitrary frame τ is then constructed as follows:

$$\begin{aligned} I_\tau(u, v, 2k-1) &= d^x_{\tau+k-1}(u, v), \\ I_\tau(u, v, 2k) &= d^y_{\tau+k-1}(u, v), \quad u = 1 \ldots w, \quad v = 1 \ldots h, \quad k = 1 \ldots L. \end{aligned} \tag{1}$$

- for an arbitrary point (u, v), the channels $I_\tau(u, v, c)$, $c = 1 \ldots 2L$ encode the motion at that point over a sequence of L frames (as illustrated in Figure 5 left)

9.2 Two-Stream CNNs

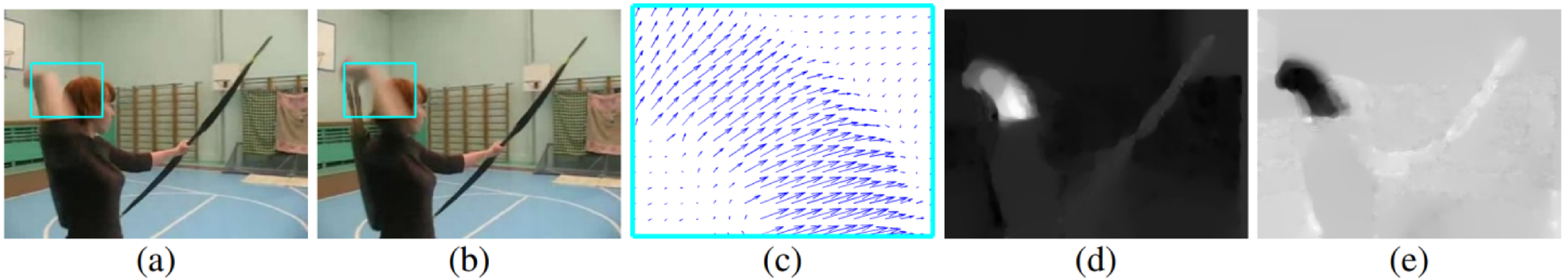


Figure 4: Optical flow. (a), (b): a pair of consecutive video frames with the area around a moving hand outlined with a cyan rectangle. (c): a close-up of dense optical flow in the outlined area; (d): horizontal component d^x of the displacement vector field (higher intensity corresponds to positive values, lower intensity to negative values). (e): vertical component d^y. Note how (d) and (e) highlight the moving hand and bow. The input to a CNN contains multiple flows.

9.2 Two-Stream CNNs

- **Trajectory stacking**: an alternative motion representation replaces the optical flow, sampled at the same locations across several frames, with the flow, sampled along the motion trajectories
- in this case, the input volume I_τ, corresponding to a frame τ, takes the following form:

$$\begin{aligned} I_\tau(u, v, 2k-1) &= d^x_{\tau+k-1}(\boldsymbol{p}_k), \\ I_\tau(u, v, 2k) &= d^y_{\tau+k-1}(\boldsymbol{p}_k), \quad u = 1 \ldots w, \quad v = 1 \ldots h, \quad k = 1 \ldots L, \end{aligned} \tag{2}$$

where $\boldsymbol{p}_k$ is the kth point along the trajectory, which starts at the location (u, v) in the frame τ and is defined by the following recurrence relation:

$$\boldsymbol{p}_1 = (u, v), \qquad \boldsymbol{p}_k = \boldsymbol{p}_{k-1} + \boldsymbol{d}_{\tau+k-2}(\boldsymbol{p}_{k-1}), \; k > 1.$$

- compared to the input volume representation (1), where the channels $I_\tau(u, v, c)$ store the displacement vectors at the locations (u, v), the input volume (2) stores the vectors sampled at the locations $\boldsymbol{p}_k$ along the trajectory (as illustrated in Figure 5 right)

9.2 Two-Stream CNNs

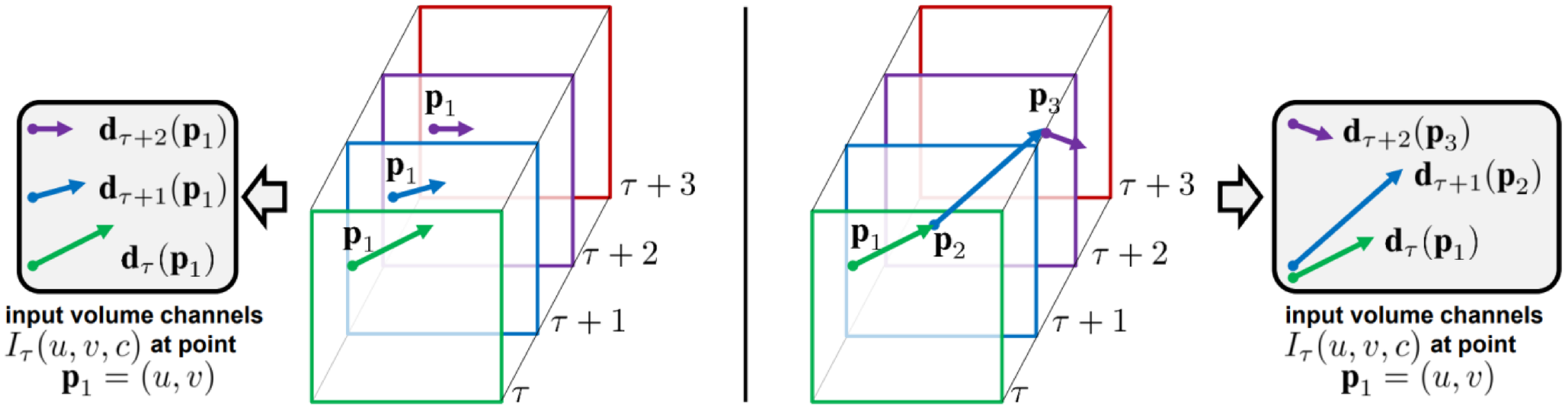


Figure 5: CNN input derivation from the multi-frame optical flow. Left: optical flow stacking (1) samples the displacement vectors ***d*** at the same location in multiple frames. Right: trajectory stacking (2) samples the vectors along the trajectory. The frames and the corresponding displacement vectors are shown with the same color.